Interaction between fluids and Helfrich membranes

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1. [?] study the steady state with arc-length gauge
2. In the dynamics, arc-length gauge does not work
3. We show how to use a generalized arc-length gauge and how to correctly incorporate it into the dynamics to obtain a proper dynamics, **??**.
4. In this dynamics
   - we implement the Crank-Nicolson method to the equations written in the reference configuration, and prove it to be numerically stable
   - we show that the derivative of mesh motion $\dot{\boldsymbol{u}}_{\rm D}$ can be obtained exactly from an auxiliary PDE withotu recurring to error-prone time discretization.
5. We couple the membrane dynamics to a bulk fluid. The membrane is a complex elastic body -¿ We proceed by steps:
   (a) Bulk fluid + rigid body **??**
   (b) Elastic body with stable elastic model (stable under compression)
   (c) Bulk fluid + elastic body with stable elastic model, **??**
   (d) Bulk fluid + membrane, **??**
   - Bulk fluid and membrane both described with Crank-Nicolson method -¿ we prove that the resulting dynamics is numerically stable. Say that you found a way to teplace $t \to t_n$ that is stable and that you could have used other ways which are also correct to within $\mathcal{O}(\Delta t)$.
   - It allows for overhangs
   - It is the first study that describes a bulk fluid coupled to a Helfrich membrane with ALE
   - It allows for turbulent behavior for both the bulk fluid and the membrane tangential flow

Perspectives:

1. Study nucleoid compaction in E. Coli with ALE
2. arc length gauge in 2d

I. FLUID AND RIGID BODY

In this Section, we will discuss the interaction between a bulk fluid and the simplest structure—a rigid body which is only allowed to turn about one point.

In all the systems that we will consider, we will introduce the reference and current configuration [? ? ?], see **??**. All quantities relative to the reference and current configuration will be denoted by the superscript reference and current, respectively. In the reference configuration, the region occupied by the bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\rm r}$. The axis of the ellipse, e..g, the rigid body, lies parallel to the $\mathsf{x}_{\rm c}^1$ axis. In the current configuration, the region occupied by the bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\rm c}$. The axis of the e'llipse, e..g, the rigid body, forms an angle θ (red arc) with respect to the $\mathsf{x}_{\rm c}^1$ axis.

The reference configuration, is related to the current one as follows [?], see **??**. At time t, the point $\mathbf{x}_{\rm c}$ corresponds to a point $\mathbf{x}_{\rm r}$ in the reference configuration through the deformation field [?]

$$\boldsymbol{u}_{\rm D}^t(\mathbf{x}_{\rm r}) \equiv \mathbf{x}_{\rm c} - \mathbf{x}_{\rm r}. \tag{1}$$

and the deformation-gradient tensor

$$F_{\alpha\beta}(\boldsymbol{u}) \equiv \delta_{\alpha\beta} + \frac{\partial u^\alpha}{\partial \mathsf{x}_{\rm r}^\beta}, \tag{2}$$

where we set

$$\partial_\alpha^{\rm r} \equiv \frac{\partial}{\partial \mathsf{x}_{\rm r}^\alpha}, \tag{3}$$

$$\partial^{\mathrm{c}}_{\alpha} \equiv \frac{\partial}{\partial \mathsf{x}^{\alpha}_{\mathrm{c}}}. \quad (4)$$

Here, we will denote the mapping between $\mathbf{x}_{\mathrm{c}}$ and $\mathbf{x}_{\mathrm{r}}$ in ?? with the fields

$$\mathbf{x}_{\mathrm{c}} = \boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), \quad (5)$$
$$\mathbf{x}_{\mathrm{r}} = \boldsymbol{\psi}^{t}(\mathbf{x}_{\mathrm{c}}). \quad (6)$$

We denote the bulk fluid velocity and surface tension by $v_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}})$ and $\sigma_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}})$, respectively, where the superscript current specifies that these fields depend on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the current configuration.

The solution of the interaction dynamics between bulk fluid and rigid body is involved because of the presence of the moving boundary $\partial\Omega^{\mathrm{c}}_{\circ}$, cf. ??. Following the arbitrary Lagrangian-Eulerian procedure, we will bridge between the Lagrangian and the Eulerian description used to describe, respectively, rigid body and bulk fluid [?]. To achieve this, in what follows we will write the equations of motion for rigid body and bulk fluid, which are naturally formulated the current configuration first, and then rewrite them in the reference configuration.

A. Rigid body motion

In this Section, we will work out the equations of motion for rigid body.

1. *Equation of motion in the current-configuration coordinates*

The dynamics of rigid body is given by the equations of motion for a rigid body which, in current coordinates, read [?]

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\int_{\partial\Omega^{\mathrm{c}}_{\circ}} \mathrm{d}l_{\mathrm{c}}\, \epsilon_{3\alpha\beta}\, \sigma^{\mathrm{c}}_{\mathrm{F}\,\beta\gamma}\hat{n}^{\mathrm{c}\,\gamma}, \quad (7)$$
$$\omega \equiv \frac{\mathrm{d}\theta}{\mathrm{d}t}. \quad (8)$$

?? relates the angular acceleration of rigid body to the torque of external forces. In what follows, greek indices $\alpha, \beta, \cdots$ will be used to denote vectors and tensors in two-dimensional Euclidean space—their position as upper or lower indices is thus immaterial [? ?]. The moment of inertia of rigid body is denoted by I. The right-hand side of ?? contains the line integral along the boundary of rigid body, of the torque of the local force exerted by bulk fluid on rigid body with respect to the left focal point $\boldsymbol{f}$ of rigid body [?]. This force is expressed in terms of the bulk fluid stress tensor [?]

$$\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta} \equiv \sigma_{\mathrm{F}\,\mathrm{c}}\delta_{\alpha\beta} + \eta_{\mathrm{F}}\left(\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\alpha}}{\partial \mathsf{x}^{\beta}_{\mathrm{c}}} + \frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\beta}}{\partial \mathsf{x}^{\alpha}_{\mathrm{c}}}\right), \quad (9)$$

where $\epsilon_{\alpha\beta\gamma}$ is the three-dimensional Levi-Civita symbol, and η_{F} the two-dimensional viscosity of bulk fluid. Finally, $\mathrm{d}l_{\mathrm{c}}$ is the line element of $\partial\Omega^{\mathrm{c}}_{\circ}$ in the current configuration, and $\hat{\boldsymbol{n}}^{\mathrm{c}}$ the unit boundary normal to $\partial\Omega^{\mathrm{c}}_{\circ}$ pointing outside Ω^{c}.

2. *Equation of motion in the reference-configuration coordinates*

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = \quad (10)$$
$$\int_{\partial\Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s\, \epsilon_{\alpha\beta}\left[\varphi^{\alpha,\,t}(\boldsymbol{\xi}(s)) - f^{\alpha}\right] \sigma^{\mathrm{r}}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s))\epsilon_{\gamma\lambda}F_{\lambda\mu}(\boldsymbol{u}^{t}_{\mathrm{D}}(\boldsymbol{\xi}(s)))\frac{\mathrm{d}\xi^{\mu}(s)}{\mathrm{d}s}$$

In ??, the integral in the right-hand side is over the curvilinear boundary $\partial\Omega^{\mathrm{r}}_{\circ}$, which is

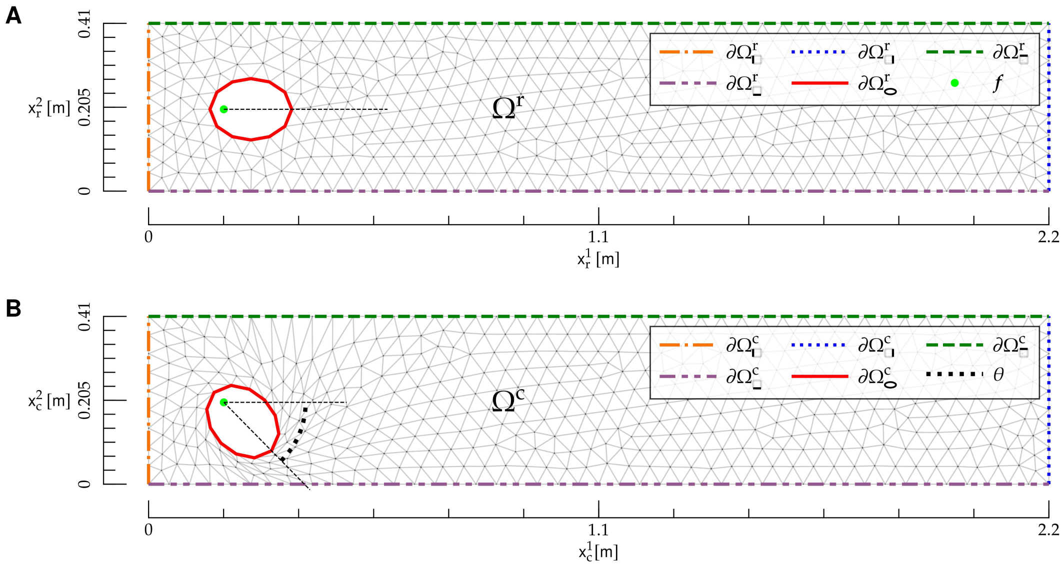

FIG. 1: Reference and current configuration for the interaction between a bulk fluid and a rigid body. **A**) Reference configuration. The region Ω^{r} is given by the rectangle $0 \leq \mathrm{x}_{\mathrm{r}}^{1} \leq L$, $0 \leq \mathrm{x}_{\mathrm{r}}^{2} \leq h$ with the elliptical hole, and is delimited by the mesh (gray lines). Boundaries in the reference configuration are denoted by $\partial\Omega^{\mathrm{r}}_{\sqsubset}$, $\partial\Omega^{\mathrm{r}}_{\sqsupset}$, $\partial\Omega^{\mathrm{r}}_{\sqcap}$, $\partial\Omega^{\mathrm{r}}_{\sqcup}$ and $\partial\Omega^{\mathrm{r}}_{\circ}$ (colored dashed lines). The ellipse focal point $\boldsymbol{f}$ (light green dot) is also shown. **B**) Current configuration. The region Ω^{c} is delimited by the mesh (gray lines). Boundaries in the current configuration are denoted by $\partial\Omega^{\mathrm{c}}_{\sqsubset}$, $\partial\Omega^{\mathrm{c}}_{\sqsupset}$, $\partial\Omega^{\mathrm{c}}_{\sqcap}$, $\partial\Omega^{\mathrm{c}}_{\sqcup}$ and $\partial\Omega^{\mathrm{c}}_{\circ}$ (colored dashed lines), and the rotational angle θ of the rigid body is shown as an arc (black dotted line). The Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$ are also shown.

parametrized as $\mathbf{x}_{\mathrm{r}} = \boldsymbol{\xi}(s)$, where s is the curvilinear coordinate. Also, $\epsilon_{\alpha\beta}$ is the two-dimensional Levi-Civita symbol [?], and $\sigma^{\mathrm{r}}_{\mathrm{F}}$ denotes the stress tensor in the reference configuration:

$$\begin{aligned}\sigma^{\mathrm{r}\,t}_{\mathrm{F}\,\alpha\beta}(\mathbf{x}_{\mathrm{r}}) \equiv& \sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}})) \qquad (11)\\ =&\sigma_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}},t)\delta_{\alpha\beta} + \eta_{\mathrm{F}}\Big[G_{\gamma\beta}(\boldsymbol{u}^{t}_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\gamma}_{\mathrm{r}}} + \\ & G_{\gamma\alpha}(\boldsymbol{u}^{t}_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}}{}^{\beta}_{\mathrm{c}}}{\partial \mathrm{x}^{\gamma}_{\mathrm{r}}}\Big],\end{aligned}$$

where

$$G_{\alpha\beta}(\boldsymbol{u}) \equiv [F(\boldsymbol{u})]^{-1}_{\alpha\beta}. \tag{12}$$

B. Fluid motion

In this Section, we will work out the equations of motion for bulk fluid.

1. Equations of motion in the current-configuration coordinates

The equations of motion, expressed in terms of the current fields $v_{\mathrm{F}\,\mathrm{c}}$ and $\sigma_{\mathrm{F}\,\mathrm{c}}$, are

$$\rho_{\mathrm{F}}\left(\partial_t v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}} + v_{\mathrm{F}}{}^{\beta}_{\mathrm{c}}\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\beta}_{\mathrm{c}}}\right) = \frac{\partial \sigma_{\mathrm{F}\,\mathrm{c}}}{\partial \mathrm{x}^{\alpha}_{\mathrm{c}}} + \eta_{\mathrm{F}}\frac{\partial}{\partial \mathrm{x}^{\beta}_{\mathrm{c}}}\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\beta}_{\mathrm{c}}}, \tag{13}$$

$$\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\alpha}_{\mathrm{c}}} = 0, \tag{14}$$

and they hold for $\mathrm{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$. **????** are the Navier-Stokes and continuity equation, respectively, in a flat, two-dimensional geometry [?]—an example of the Navier-Stokes equations on curved geometry will be presented in X. There, ρ_{F} is the two-dimensional bulk fluid density.

The boundary conditions for **????????** are

$$v_{\mathrm{F}}{}^{1}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = v_{*} \;\text{ on } \partial\Omega^{\mathrm{c}}_{\sqsubset}, \tag{15}$$

$$v_{\mathrm{F}}{}^{2}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = 0 \;\text{ on } \partial\Omega^{\mathrm{c}}_{\sqsubset}, \tag{16}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \mathbf{0} \;\text{ on } \partial\Omega^{\mathrm{c}}_{\sqcup}, \tag{17}$$

$$v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathbf{x}_{\mathrm{c}}) - f^{\beta}] \;\text{ on } \partial\Omega^{\mathrm{c}}_{\circ}, \tag{18}$$

$$\eta_{\mathrm{F}} \frac{\partial {v_{\mathrm{F}}}_{\mathrm{c}}^{\alpha}}{\partial \mathrm{x}_{\mathrm{c}}^{1}} = 0 \;\; \text{on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{19}$$

$$\sigma_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = 0 \;\; \text{on } \partial\Omega^{\mathrm{c}}_{\square}. \tag{20}$$

???? ensure that bulk fluid is injected on the boundary $\partial\Omega^{\mathrm{c}}_{\square}$ in a direction parallel to the $\mathrm{x}_{\mathrm{c}}^{1}$ axis, and **??** is a no-slip boundary condition [?] along the channel walls $\partial\Omega^{\mathrm{c}}_{=}$, see **??**. **??** ensures that, at the boundary $\partial\Omega^{\mathrm{c}}_{\circ}$, the bulk fluid velocity matches the rotational velocity of rigid body. Given that rigid body can only rotate about its left focal point $\boldsymbol{f}$, the rigid body rotational velocity of a point $\mathbf{x}_{\mathrm{c}} \in \partial\Omega^{\mathrm{c}}_{\circ}$ is given by

$$\frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t} [\psi^{t\,\beta}(\mathbf{x}_{\mathrm{c}}) - c_{\beta}], \tag{21}$$

where $\boldsymbol{R}$ is the rotation matrix corresponding to a counterclockwise rotation:

$$\boldsymbol{R}(\theta) \equiv \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{22}$$

Finally, **????** ensure, respectively, that on $\partial\Omega^{\mathrm{c}}_{\square}$ there is zero traction and tension; this corresponds to the hypothesis that there is free flow at $\partial\Omega^{\mathrm{c}}_{\square}$ [? ?].

2. *Equations of motion in the reference-configuration coordinates*

For each field $\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}), \sigma_{\mathrm{F}}(\mathbf{x}_{\mathrm{c}})$ which depends on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the current configuration, we introduce its counterpart in the reference configuration:

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t), \tag{23}$$

$$\sigma_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{\sigma}_{\mathrm{F}\,\mathrm{r}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t). \tag{24}$$

We will now rewrite the equations of motion and the boundary conditions in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}$, $\sigma_{\mathrm{F}\,\mathrm{c}}$, and the reference coordinate $\mathbf{x}_{\mathrm{r}}$. The equations of motion **????** yield

$$\rho_{\mathrm{F}} \left\{ \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{\alpha}(\mathbf{x}_{\mathrm{r}}, t)}{\partial t} + \left[{v_{\mathrm{F}}}_{\mathrm{r}}^{\gamma}(\mathbf{x}_{\mathrm{r}}, t) - \frac{\mathrm{d}\varphi^{\gamma\,t}(\mathbf{x}_{\mathrm{r}})}{\mathrm{d}t} \right] G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{\alpha}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} \right\} = \tag{25}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial \sigma_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} + \eta_{\mathrm{F}}\, G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial}{\partial \mathrm{x}_{\mathrm{r}}^{\delta}} \left[G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial v_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathrm{x}_{\mathrm{r}}^{\gamma}} \right],$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{\alpha}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} = 0, \tag{26}$$

which hold in Ω^{r}.

The boundary conditions **????????????** are rewritten as

$${v_{\mathrm{F}}}_{\mathrm{r}}^{1}(\mathbf{x}_{\mathrm{r}}) = v_{*} \;\; \text{on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{27}$$

$${v_{\mathrm{F}}}_{\mathrm{r}}^{2}(\mathbf{x}_{\mathrm{r}}) = 0 \;\; \text{on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{28}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}) = \mathbf{0} \;\; \text{on } \partial\Omega^{\mathrm{r}}_{=}, \tag{29}$$

$${v_{\mathrm{F}}}_{\mathrm{r}}^{\alpha}(\mathbf{x}_{\mathrm{r}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t} [\mathrm{x}_{\mathrm{r}}^{\beta} - f^{\beta}] \;\; \text{on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{30}$$

$$\eta_{\mathrm{F}} G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{\alpha}}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} = 0 \;\; \text{on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{31}$$

$$\sigma_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}) = 0 \;\; \text{on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{32}$$

C. Motion of bulk fluid domain

The equations of motion **??????????????????** involve the displacement field $\boldsymbol{u}_{\mathrm{D}}$, which describes how the region—or domain—which contains bulk fluid, is deformed in time, cf. **??**. We will denote this region as the bulk fluid domain. We observe that this field has, so far, been left arbitrary, and we have the freedom to choose it.

Here, $\boldsymbol{u}_{\mathrm{D}}$ will be determined with an analogy with the theory of elasticity [?]. We imagine that, as rigid body rotates about its focal point, each point $\mathbf{x}_{\mathrm{r}} \in \Omega^{\mathrm{r}}$ moves into a point $\mathbf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$, and that the domain Ω^{r} is deformed, e.g., locally stretched and compressed, into Ω^{c}, like an elastic medium.

We will thus determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem given by the following elastic model [? ?]

$$\frac{\partial S_{\mathrm{D}\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{33}$$

where the elastic stress tensor S_{D} is given by

$$S_{\mathrm{D}\alpha\beta} \equiv F_{\alpha\gamma} T_{\gamma\beta}, \tag{34}$$

and

$$T_{\alpha\beta} \equiv K(\boldsymbol{u}_{\mathrm{D}}) E_{\gamma\gamma}\delta_{\alpha\beta} + 2\mu(\boldsymbol{u}_{\mathrm{D}}) \left[E_{\alpha\beta} - \frac{1}{2}\delta_{\alpha\beta} E_{\gamma\gamma} \right], \tag{35}$$

$$E_{\alpha\beta} \equiv \frac{1}{2}\left(F_{\gamma\alpha}F_{\gamma\beta} - \delta_{\alpha\beta}\right), \tag{36}$$

$$K(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}, \tag{37}$$

$$\mu(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}. \tag{38}$$

In ????, we included an additional dependence of bulk modulus K and the modulus of compression μ on the deformation-gradient tensor ??. In the absence of this dependence, the elastic model above would be unstable under local compression. In fact, the model's Lagrangian would not penalize configurations with small $\det F$. The inclusion of the dependence ???? on $\det F$ allows for penalizing these configurations—the larger the exponent $\zeta > 0$, the stronger the penalty. As a result, this dependence makes the model stable under mesh compression, such as the one shown below rigid body in ??B.

The boundary conditions for the partial differential equation ?? are

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{39}$$

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{f} - \mathsf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta)\cdot(\mathsf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}. \tag{40}$$

?? enforces the fact that the mesh is not deformed at the boundary $\partial\Omega^{\mathrm{r}}_{\square}$. Finally, ?? ensures that the mesh deformation at $\partial\Omega^{\mathrm{r}}_{\circ}$ matches the deformation induced by the rotation of rigid body.

?????? constitute the boundary-value problem which determines $\boldsymbol{u}_{\mathrm{D}}$. Along the lines of boundary-value problems in the theory of elasticity, such boundary-value problem is already, naturally formulated in the reference configuration [?].

From the boundary-value problem above, we obtain a boundary-value problem for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, which will be needed in the following to describe the bulk fluid domain motion. By deriving both sides of ??????, we obtain, respectively,

$$\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} \frac{\mathrm{d}S_{\mathrm{D}\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{41}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{42}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\cdot(\mathsf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{43}$$

where in ???? we used the fact that x_{r} is independent of time, and in ?? we substituted ??. The left-hand side of ?? depends on the second partial derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$—see ??????????. As a result, given θ, ω, and $\boldsymbol{u}_{\mathrm{D}}$ from ??????, the boundary-value problem ?????? yields $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

Overall ?????????????????????????????????? constitute a boundary-value problem for $\theta, \omega, \boldsymbol{v}_{\mathrm{F\,r}}, \sigma_{\mathrm{F\,r}}$, $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$ which, combined with the temporal boundary conditions

$$\theta(t=0) = \theta_0 \tag{44}$$

$$\omega(t=0) = \omega_0 \tag{45}$$

$$\boldsymbol{v}_{\mathrm{F\,r}}(\mathsf{x}_{\mathrm{r}}, t=0) = \boldsymbol{v}_{\mathrm{F\,r}0}(\mathsf{x}_{\mathrm{r}}), \tag{46}$$

$$\boldsymbol{\sigma}_{\mathrm{F\,r}}(\mathsf{x}_{\mathrm{r}}, t=0) = \boldsymbol{\sigma}_{\mathrm{F\,r}0}(\mathsf{x}_{\mathrm{r}}), \tag{47}$$

determine the dynamics of the system.

D. Temporal discretization

To numerically solve for the dynamics in a time span $0 \le t \le T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\,\Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson discretization method for Navier-Stokes equations [? ?] with the arbitrary Lagrangian-Eulerian formulation, by discretizing the dynamical equations for rigid body, bulk fluid and bulk fluid domain—see below.

1. *Rigid body*

Proceeding along the Crank Nicolson scheme [?], we define fields at staggered, integer and semi-integer time steps scheme [?]. We set

$$\sigma_{\mathrm{F\,r}}{}^{n}(\mathsf{x}_{\mathrm{r}}) \equiv \sigma_{\mathrm{F\,c}}(\mathsf{x}_{\mathrm{r}}, t_n) \tag{48}$$

$$\sigma_{\mathrm{F\,r}}{}^{n-1/2}(\mathsf{x}_{\mathrm{r}}) \equiv \sigma_{\mathrm{F\,c}}\left(\mathsf{x}_{\mathrm{r}}, \frac{t_n + t_{n-1}}{2}\right), \tag{49}$$

and similarly for other quantities.

We discretize ???? and, neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\frac{\theta^n - \theta^{n-1}}{\Delta t} = \omega^n, \tag{50}$$

$$\frac{\omega^n - \omega^{n-1}}{\Delta t} = I \int_{\partial \Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s \, \epsilon_{\alpha\beta} \left[\xi^\alpha(s) + u^{n-1,\,\alpha}(\boldsymbol{\xi}(s)) - f^\alpha \right] \sigma^{\mathrm{r}\, n-1}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s); {\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}, {\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-3/2}) \times \epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}_{\mathrm{D}}^{n-1}(\boldsymbol{\xi}(s))) \frac{\mathrm{d}\xi^\mu(s)}{\mathrm{d}s}, \tag{51}$$

where in ?? we used ???? and we explicitly indicated the dependence of the stress tensor ?? on ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{c}}$ and ${\sigma_{\mathrm{F}}}_{\mathrm{c}}$.

2. Bulk fluid

To discretize in time the bulk fluid problem, we will apply the Crank Nicolson splitting scheme [?] to the Navier-Stokes equations ???? in the reference configurations. This splitting scheme is derived form the incremental pressure correction scheme [?], in which the solution of the Navier-Stokes equations is split into three intermediate steps. In what follows, we will synonymously use the terms 'pressure' and 'surface tension' for conciseness [?].

The discrete form of the partial differential equations ???? is

$$\rho_{\mathrm{F}} \left[\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} - {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t} + \frac{3}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma} - \dot{u}^{n-1,\,\gamma})\, G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} - \frac{1}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma} - \dot{u}^{n-2,\,\gamma}) \times G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} \right] = G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial {\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} + \eta_{\mathrm{F}}\, G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\delta}} \left[G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}} \left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} + {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2} \right) \right], \tag{52}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} = 0, \tag{53}$$

where we omitted the dependence on $\mathbf{x}_{\mathrm{r}}$ for clarity. Also, in the left-hand side of ?? we rewrote the convective term according to the Adams-Bashforth discretization scheme [? ?], and we used ???? to rewrite the time derivative of $\boldsymbol{\varphi}$ in terms of $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

The discrete form of the boundary conditions ???????????? reads

$${v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,1} = v_* \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{54}$$

$${v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,2} = 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{55}$$

$${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n} = \boldsymbol{0} \;\text{ on } \partial\Omega^{\mathrm{r}}_{=}, \tag{56}$$

$${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}} = \omega^n \left. \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta} \right|_{\theta=\theta_n} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{57}$$

$$\eta_{\mathrm{F}}\, G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}+{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2}\right)=0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{58}$$

$${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2}=0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{59}$$

We will now split the boundary-value problem into three steps [? ?]:

1. **Approximated velocity.** We introduce the velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$, which approximates the true velocity ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}$, and which satisfies the boundary-value problem

$$\rho_{\mathrm{F}}\left\{\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t}+\left[\frac{3}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-1,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1})-\frac{1}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-2,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2})\right]\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right\}= \tag{60}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial {\sigma_{\mathrm{F}}}_{\mathrm{r}}^{*}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}+\eta_{\mathrm{F}}\,G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\delta}}\left(G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\right),$$

$$\overline{v}_{\mathrm{F}}^{1}=v_{*} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{61}$$

$$\overline{v}_{\mathrm{F}}^{2}=0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{62}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\boldsymbol{0} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{63}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\omega^{n}\left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n}\cdot(\mathbf{x}_{\mathrm{r}}-\boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{64}$$

$$\eta_{\mathrm{F}}\,G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}=0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{65}$$

where

$$\boldsymbol{V}_{\mathrm{F}}\equiv\frac{\overline{\boldsymbol{v}}_{\mathrm{F}}+{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}}{2}, \tag{66}$$

$${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{*}\equiv{\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-3/2}. \tag{67}$$

2. **Pressure correction.**

Subtracting **????** and neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\rho_{\mathrm{F}}\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\Delta t}=G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial\phi}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} \text{ in } \Omega^{\mathrm{r}}, \tag{68}$$

where

$$\phi\equiv{\sigma_{\mathrm{F}}}_{\mathrm{c}}^{*}-{\sigma_{\mathrm{F}}}_{\mathrm{c}}^{n-1/2} \tag{69}$$

is the surface-tension increment. By applying $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\partial/\partial \mathsf{x}_{\mathrm{r}}^{\gamma}$ to both sides of **??** and using **??**, we obtain a second-order partial differential equation for ϕ:

$$\frac{\rho_{\mathrm{F}}}{\Delta t}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial\overline{v}_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}= \tag{70}$$

$$G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\left(G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right).$$

We will now work out the boundary conditions for **??**. **??????** imply the boundary condition

$$\phi=0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{71}$$

Multiplying **??** by $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}$, where $\hat{\boldsymbol{n}}^{\mathrm{r}}$ is the unit boundary normal in the reference configuration, we obtain

$$\frac{\rho_{\mathrm{F}}}{\Delta t}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}(\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha})= \tag{72}$$

$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}.$$

Combining **??** with **????????????????**, we obtain the second boundary condition

$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}= \tag{73}$$

$$0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\sqsubset} \cup \partial\Omega^{\mathrm{r}}_{\circ}.$$

The boundary-value problem ?????? determines ϕ and, through the definition ??, the surface tension ${\sigma_{\mathrm{c}}}^{n-1/2}$.

3. **Velocity correction.** The velocity ${\boldsymbol{v}_{\mathrm{F}_{\mathrm{r}}}}^{n}$ is determined from the approximate velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$ from ??.

3. *Bulk fluid domain*

The discrete version of the boundary-value problem for $\boldsymbol{u}_{\mathrm{D}}$, ??????, reads

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}^n_{\mathrm{D}})}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{74}$$

$$\boldsymbol{u}^n_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{75}$$

$$\boldsymbol{u}^n_{\mathrm{D}} = \boldsymbol{f} - \mathsf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta^n)\cdot(\mathsf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{76}$$

and that of the boundary-value problem for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, ??????, is

$$\frac{\partial}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}}\frac{\mathrm{d}S_{\mathrm{D}\,\alpha\beta}(\dot{\boldsymbol{u}}^n_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{77}$$

$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{78}$$

$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \omega\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\cdot(\mathsf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{79}$$

We can now iterate in time for the ensemble of fields as follows. At each step n, given θ^{n-1}, ω^{n-1}, ${\boldsymbol{v}_{\mathrm{F}_{\mathrm{r}}}}^{n-1}$, ${\sigma_{\mathrm{F}_{\mathrm{r}}}}^{n-3/2}$, $\boldsymbol{u}^{n-1}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^{n-1}_{\mathrm{D}}$ from the preceding step, we

1. **Update rigid body.** Solve the algebraic equations ???? for θ^n and ω^n.
2. **Update bulk fluid domain.** Solve the boundary-value problems ???????????? for $\boldsymbol{u}^n_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^n_{\mathrm{D}}$.
3. **Update bulk fluid.** Solve the boundary-value problems ???????????? and ?????? for $\overline{\boldsymbol{v}}_{\mathrm{F}}$ and ${\sigma_{\mathrm{F}_{\mathrm{r}}}}^{n-1/2}$, and obtain $\boldsymbol{v}^n_{\mathrm{F}_{\mathrm{r}}}$ from ??.

The fields ${\boldsymbol{v}_{\mathrm{F}_{\mathrm{r}}}}^{n-1}$, ${\sigma_{\mathrm{F}_{\mathrm{r}}}}^{n-3/2}$ in the current configuration are then obtained from the respective fields in the reference configuration through ????.

An example of such dynamics is shown in ??. The channel has dimensions $L = 2.2\,\mathrm{m}$, $h = 0.41\,\mathrm{m}$, cf. ??, $\rho_{\mathrm{F}} = 1\,\mathrm{Kg/m^2}, \eta_{\mathrm{F}} = 10^{-3}\,\mathrm{Pa\,m\,sec}$, and the inflow velocity profile is a Poiseuille flow $v_*(\mathsf{x}_{\mathrm{c}}) = 6v_0\mathsf{x}^2_{\mathrm{c}}(\mathsf{x}^2_{\mathrm{c}} - h)/h^2$, with $v_0 = 1\,\mathrm{m/sec}$. These parameters have been taken from the FEAT2D DFG 2D-3 benchmark for a flow around a cylinder [?]. The bulk fluid domain mesh stiffness exponent has been chosen to be $\zeta = 3$ [?]. Finally, rigid body is an ellipse with semi-axes $a = 0.1\,\mathrm{m}$, $b = 0.075\,\mathrm{m}$, center located at $\mathsf{x}_{\mathrm{r}} = (0.25\,\mathrm{m}, 0.2\,\mathrm{m})$, and moment of inertia $I = 10^{-1}\,\mathrm{Kg\,m^2}$. These parameters correspond approximately to a light gas interacting with a solid body made of light wood.

As shown in ??, the dynamics of ?? involves two physical time scales: A short one, τ_{S}, related to fast angular oscillations of body, and a long one, τ_{L}, given by the damping of such oscillations.

First, the scale τ_{S} can be estimated by considering the dynamical equation ?? for body, and approximating its left-hand side as I/τ^2_{S} and its right-hand side as $(\eta_{\mathrm{F}}v_0/a\,2\pi a)a$. Here, $\eta_{\mathrm{F}}v_0/a$ is the viscous component of the force per unit length exerted by bulk fluid on body, cf. ??, $2\pi a$ estimates the total length of the boundary of body, and a approximates the lever arm of the force above. Putting everything together, we obtain

$$\tau_{\mathrm{S}} \sim \sqrt{\frac{I}{2\pi a\eta_{\mathrm{F}}v_0}}. \tag{80}$$

For the parameters of ??, ?? yields $\tau_{\mathrm{S}} \sim 13$ sec, which is in agreement with the period of the oscillations in ??.

Second, τ_{L} can be estimated by building a combination of the material parameters ρ_{F}, η_{F} and I of the model. The only parameter combination that can yield a quantity with the dimensions of a time is

$$\tau_{\mathrm{L}} \sim \frac{\sqrt{\rho_{\mathrm{F}}I}}{\eta_{\mathrm{F}}}. \tag{81}$$

For the parameters of ??, ?? yields $\tau_{\mathrm{L}} \sim 316$ sec, which is roughly the timescale at which the width of the oscillations in ?? is dampened out.

II. FLUID AND ELASTIC BODY

In this Section, we will build on the analysis of ?? and put bulk fluid into interaction with a more complex physical object: an elastic body. Only the

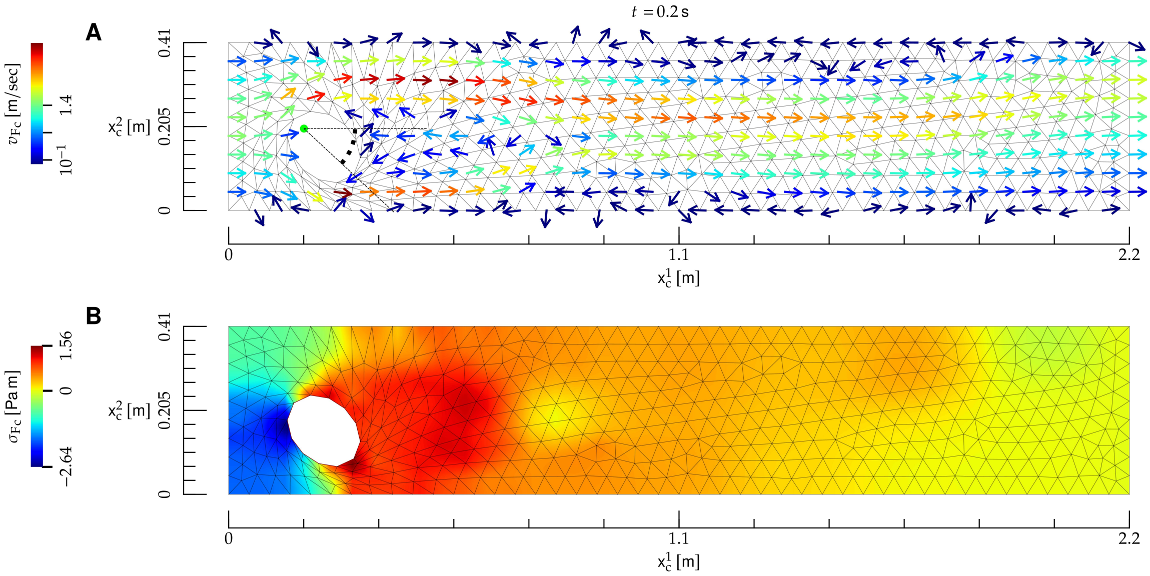

FIG. 2: Interaction between a bulk fluid and a rigid body, whose material properties roughly correspond to a light gas and a body made of light wood, respectively. **A**) Temporal snapshot of the mesh (black lines) and velocity field (arrows) at the current time t, shown on top. The direction of the velocity field is indicated by the arrows, and its norm by their color; the direction of arrows with velocity close to zero is not defined. The body (ellipse) is allowed to pivot about its left focal point (red dot), and the related angle with respect to the x_c^1 direction is denoted by a red arc. **B**) Color map of the surface tension at the current instant of time t.

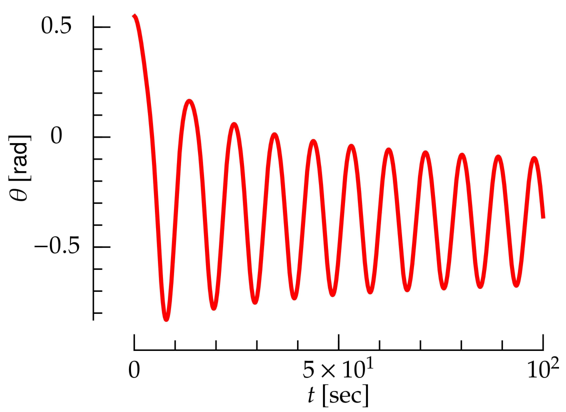

FIG. 3: Angle of the body as a function of time for the interaction between a bulk fluid and a rigid body shown in **??**.

main results will be presented here; their derivation follows the lines of **??**.

The reference and current configurations are shown in **??**. Here, the elastic body boundary $\partial\Omega_{\bigcirc}$ may deform, but the inner elastic body boundary $\partial\Omega_{\odot}$ is pinned to the x_c^1, x_c^2 plane.

The equations of motion are the following.

First, the dynamics of bulk fluid is governed by the Navier-Stokes and mass-conservation equations **????**, respectively. Their boundary conditions are **??????????**, which stay unchanged. On the other hand, boundary condition **??**, which ensures that the bulk fluid velocity matches the body velocity at the interface between bulk fluid and body, now reads

$$\boldsymbol{v}_{\mathrm{F\,c}}(\mathbf{x}_\mathrm{c}) = \left.\frac{\mathrm{d}\boldsymbol{u}_\mathrm{E}(\mathbf{x}_\mathrm{r})}{\mathrm{d}t}\right|_{\mathbf{x}_\mathrm{r}=\psi^t(\mathbf{x}_\mathrm{c})} \quad \text{on } \partial\Omega^\mathrm{c}_{\bigcirc} \tag{82}$$

Second, the motion of elastic body is governed by Newton's equations of motion for an elastic body [**?**]

$$\rho_\mathrm{E}\frac{\mathrm{d}^2 u^\alpha_\mathrm{E}(\mathbf{x}_\mathrm{r})}{\mathrm{d}t^2} = \frac{\partial S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}_\mathrm{E})}{\partial \mathrm{x}^\beta_\mathrm{r}}, \tag{83}$$

where ρ_E is the density of elastic body which is assumed to be independent of space. Also, $\boldsymbol{u}_\mathrm{E}$ is the deformation field of elastic body, which is defined along the lines of **??**. The elastic body stress tensor is derived from a compressible neo-Hookean model [**?**], and it reads

$$S_{\mathrm{E}\alpha\beta} \equiv \frac{\mu}{\det(F)}\left(-\frac{1}{2}C_{\gamma\gamma}G_{\alpha\beta} + F_{\beta\alpha}\right) + K[(\det(F))^2 - 1]G_{\alpha\beta}, \tag{84}$$

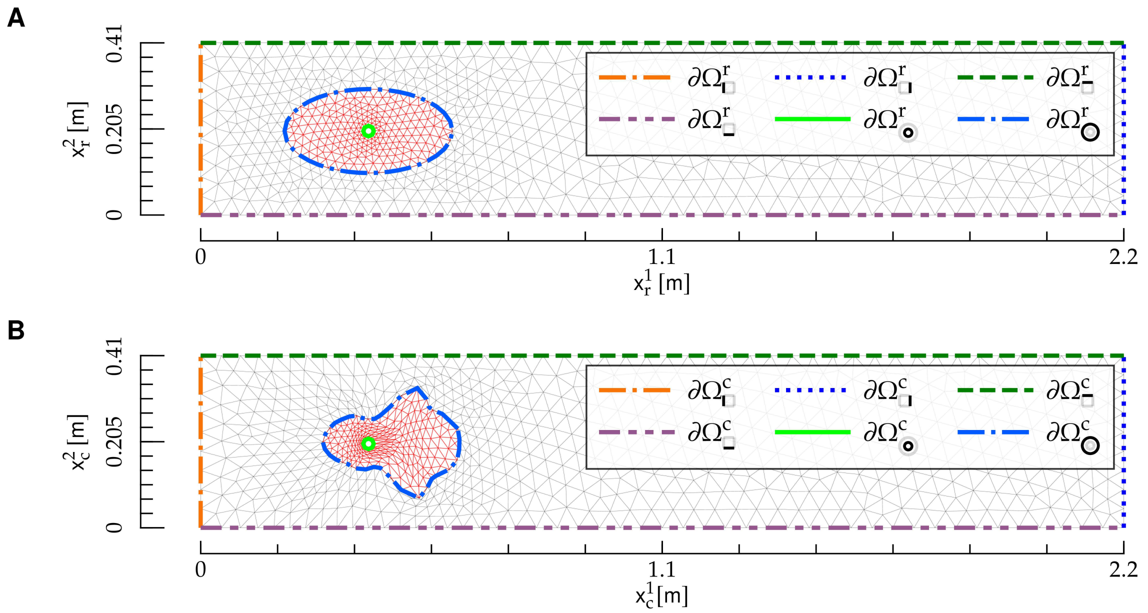

FIG. 4: Reference and current configuration for the interaction between a bulk fluid and an elastic body. The figures follows the same notation as ??. In panels **A** and **B**, the domains $\Omega^{\rm r}$ and $\Omega^{\rm c}$ are the regions covered by the mesh, respectively.

where

$$C_{\alpha\beta} \equiv \delta_{\alpha\beta} + 2E_{\alpha\beta}. \tag{85}$$

We have chosen this model because it is stable under compression, i.e., its potential energy diverges as $\det(F) \to 0$ [?]. As opposed to linear models [?] and to the elastic model of ??, this model proves to yield a stable dynamics for elastic body as it is compressed by bulk fluid, see below and ??. The boundary conditions for ?? are

$$\boldsymbol{u}_{\rm E}(\mathbf{x}_{\rm r}) = 0 \text{ on } \partial\Omega^{\rm r}_{\odot}, \tag{86}$$

$$\epsilon_{\beta\gamma}\; S_{{\rm E}\,\alpha\beta}(\boldsymbol{u}^t_{\rm E})\big|_{\mathbf{x}_{\rm r}=\boldsymbol{\xi}(s)} \frac{{\rm d}\xi^\gamma}{{\rm d}s} = \tag{87}$$

$$\sigma^{\rm c}_{{\rm F}\,\alpha\beta}(\boldsymbol{\varphi}^t(\boldsymbol{\xi}(s)))\epsilon_{\beta\gamma}\frac{{\rm d}\varphi^{t,\gamma}(\boldsymbol{\xi}(s))}{{\rm d}s} \text{ on } \partial\Omega^{\rm r}_{\bigcirc}.$$

Finally, the equations of motion for bulk fluid domain are ????, with boundary conditions ???? and

$$\boldsymbol{u}^t_{\rm D}(\mathbf{x}_{\rm r}) = \boldsymbol{u}^t_{\rm E}(\mathbf{x}_{\rm r}) \text{ on } \partial\Omega^{\rm r}_{\bigcirc}, \tag{88}$$

$$\dot{\boldsymbol{u}}^t_{\rm D}(\mathbf{x}_{\rm r}) = \dot{\boldsymbol{u}}^t_{\rm E}(\mathbf{x}_{\rm r}) \text{ on } \partial\Omega^{\rm r}_{\bigcirc}. \tag{89}$$

?? and ?? ensure that the displacement and velocity of elastic body and bulk fluid domain are conforming at the elastic body-bulk fluid domain interface, and they replace ????, respectively.

The system dynamics is obtained as follows: given ${\boldsymbol{v}_{\rm F}}_{\rm r}{}^{n-1}$, ${\sigma_{\rm F}}_{\rm r}{}^{n-3/2}$, $\boldsymbol{u}^{n-1}_{\rm E}$, $\dot{\boldsymbol{u}}^{n-1}_{\rm E}$, $\dot{\boldsymbol{u}}^{n-1}_{\rm D}$ and $\dot{\boldsymbol{u}}^{n-1}_{\rm D}$ from the preceding step, we

- **Update elastic body.** Solve for $\boldsymbol{u}^n_{\rm E}$ and $\dot{\boldsymbol{u}}^n_{\rm E}$ the discrete version of ??

$$\rho_{\rm E}\frac{\dot{u}^{n,\,\alpha}_{\rm E} - \dot{u}^{n-1,\,\alpha}_{\rm E}}{\Delta t} = \frac{\partial S_{{\rm E}\,\alpha\beta}(\boldsymbol{u}^n_{\rm E})}{\partial {\rm x}^\beta_{\rm r}}, \tag{90}$$

$$\frac{\boldsymbol{u}^n_{\rm E} - \boldsymbol{u}^{n-1}_{\rm E}}{\Delta t} = \dot{\boldsymbol{u}}_{\rm E}, \tag{91}$$

with boundary conditions obtained from ????:

$$\boldsymbol{u}^n_{\rm E} = 0 \text{ on } \partial\Omega^{\rm r}_{\odot}, \tag{92}$$

$$\epsilon_{\beta\gamma}\;\; S_{{\rm E}\,\alpha\beta}(\boldsymbol{u}^n_{\rm E})\big|_{\mathbf{x}_{\rm r}=\boldsymbol{\xi}(s)} \frac{{\rm d}\xi^\gamma}{{\rm d}s} = \tag{93}$$

$$\sigma^{\mathrm{r}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\xi}(s); {\boldsymbol{v}_{\mathrm{F}}}^{n-1}_{\mathrm{r}}, {\sigma_{\mathrm{F}}}^{n-3/2}_{\mathrm{r}}, \boldsymbol{u}^n_{\mathrm{E}})\epsilon_{\beta\gamma}\, F_{\gamma\delta}(\boldsymbol{u}^{n-1}_{\mathrm{E}})\big|_{\mathsf{x}_{\mathrm{r}}=\boldsymbol{\xi}(s)} \frac{\mathrm{d}\xi^\delta}{\mathrm{d}s} \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}.$$

In ??, we have used ???? to rewrite the derivative of $\boldsymbol{\phi}(\boldsymbol{\xi}(s))$.

- **Update bulk fluid domain.** Solve for $\boldsymbol{u}^n_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^n_{\mathrm{D}}$ the boundary-value problems given by ???? with boundary conditions ??, ?? and

$$\boldsymbol{u}^n_{\mathrm{D}} = \boldsymbol{u}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}, \tag{94}$$

$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \dot{\boldsymbol{u}}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}. \tag{95}$$

- **Update bulk fluid.** Solve the boundary-value problems given by ???? and boundary conditions ???????????? and

$$\overline{\boldsymbol{v}}_{\mathrm{F}} = \dot{\boldsymbol{u}}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}, \tag{96}$$

  which replaces ?? for a rigid body, and solve ??.

An example of the coupled dynamics of bulk fluid and elastic body is shown in ??. Here, $\partial\Omega^{\mathrm{r}}_{\bigcirc}$ is an ellipse with semi-axes $a = 0.2\,\mathrm{m}$, $b = 0.1\,\mathrm{m}$ and center located at $\mathsf{x}_{\mathrm{r}} = (0.4\,\mathrm{m}, 0.2\,\mathrm{m})$, and $\partial\Omega^{\mathrm{r}}_{\odot}$ has radius $r = 0.0125\,\mathrm{m}$ and center located at x_{r}. The elastic body density and elastic moduli are, respectively, $\rho_{\mathrm{E}} = 10^3\,\mathrm{Kg/m^2}$ and $K = \mu = 10\,\mathrm{Kg/\,sec^2}$. The other parameters are the same as in ??.

III. FLUID AND HELFRICH MEMBRANE

In this Section, we will build on the analysis of ??, and complexify the structure of the elastic body with which bulk fluid interacts by considering, instead of an elastic material, an Helfrich membrane [? ?].

For this problem, we will consider the geometry depicted in ??. A vertical channel with mirror symmetry about its mid axis is considered. The left boundary of the channel is $\partial\Omega_{\square}$, and the boundary $\partial\Omega_{\square}$ lies on the symmetry axis. Fluid is injected from bottom to top on $\partial\Omega_{\square}$. Helfrich membrane lies on the top boundary of bulk fluid, $\partial\Omega_{\square}$, which coincides with the Helfrich membrane manifold $\partial\Omega_{\_\_}$. Finally, we denote by Ω the region enclosed by the boundaries, and all the quantities above will be considered in the reference and current configurations.

The geometry of ?? may represent bulk fluid being confined in a large reservoir, lying on bottom of the channel, where the channel is a vertical neck of the reservoir, covered on top by Helfrich membrane.

A. Helfrich membrane motion

Both the physical nature and mathematical description of Helfrich membrane is significantly more complex than that of elastic body. First, unlike elastic body, Helfrich membrane has a fluid structure: its material elements can flow both tangentially and normally to it. Such fluid behavior requires an Eulerian description, which then needs to be connected to the Lagrangian description used to describe bulk fluid domain. Second, Helfrich membrane is described by fourth-order partial differential equations [?], which are significantly more complex than the second-order partial differential equations which describe elastic body [?]. Third, the geometrical description of Helfrich membrane requires fixing an appropriate *gauge*. In fact, a general parametrization of Helfrich membrane is given by two functions $\mathsf{x}^{\mathrm{M}}_{\mathrm{c}}(x^1)$. Both of these functions depend on the curvilinear coordinate x^1 which is defined in the reference domain of Helfrich membrane, $\partial\Omega^{\mathrm{r}}_{\_\_}$, see ??—note that here x^i differs from the Cartesian coordinate x^α. Despite this pair of functions, a single partial differential equation determines the Helfrich membrane shape [?]. To understand this redundancy, consider the stretching field $\nu > 0$, defined by

$$\nu^2 \equiv \boldsymbol{e}_1 \cdot \boldsymbol{e}_1, \tag{97}$$

$$\boldsymbol{e}_i \equiv \frac{\partial \mathsf{x}^{\mathrm{M}}_{\mathrm{c}}}{\partial x^i}. \tag{98}$$

It is then easy to realize that two parametrizations with different νs may yield the same physical shape. Such redundancy may be removed by a gauge fixing, i.e., by adding a constraint on $\mathsf{x}^{\mathrm{M}}_{\mathrm{c}}$, or its derivatives. Two common gauge choices are the Monge [? ?], or the arc-length parametrization [?], in which one sets

$$\nu = 1. \tag{99}$$

The arc-length gauge has been used to describe, for instance, the steady state for membrane tubules [?]. However, while a static constraint such as ?? is suited to static shapes, it cannot describe the Helfrich membrane dynamics, such as the one

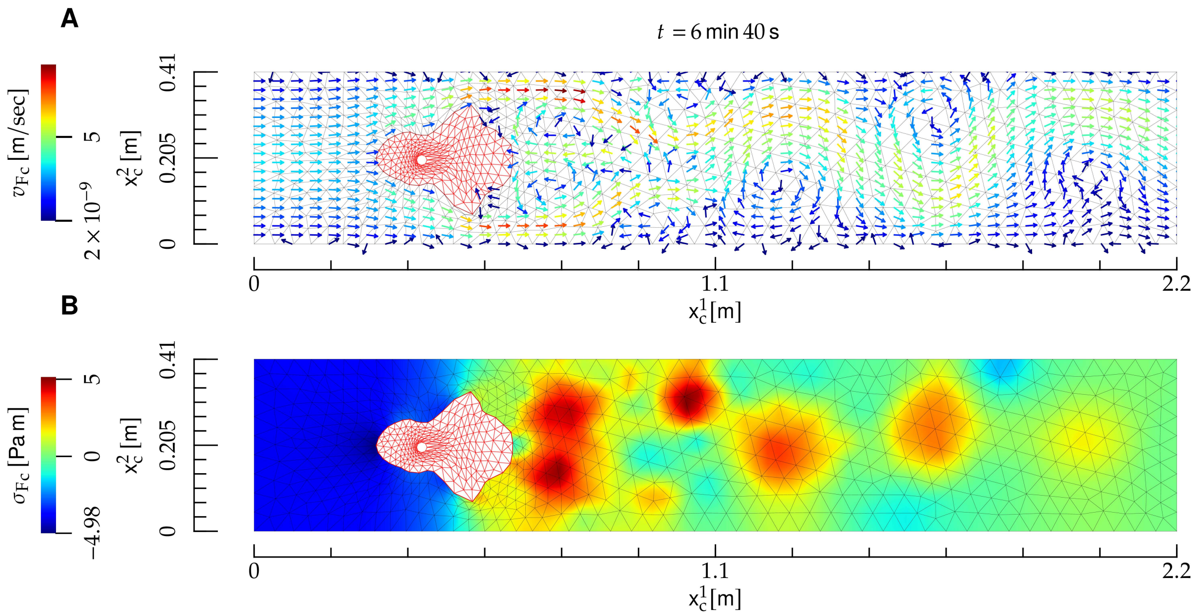

FIG. 5: Interaction between a bulk fluid and an elastic body. The notation is the same as in ??, and the elastic body is depicted as a red mesh.

of Helfrich membrane interacting with bulk fluid. In fact, as Helfrich membrane is deformed, it will also stretch: as a result, ?? must be replaced by a time-dependent gauge—see below.

We describe Helfrich membrane with the Lagrangian approach, along the lines of the description of bulk fluid domain in ?? and elastic body in ??. The Helfrich membrane displacement field is

$$\boldsymbol{u}_{\rm M}(x^1) \equiv \mathbf{x}_{\rm c}^{\rm M}(x^1) - \mathbf{x}_{\rm r}^{\rm M}(x^1), \tag{100}$$

where we choose the reference configuration as a straight line

$$\mathbf{x}_{\rm r}^{\rm M}(x^1) \equiv (x^1, h),\, 0 \leq x^1 \leq L, \tag{101}$$

which is shown in ??A. In addition, we introduce the angle ψ formed by the tangent to Helfrich membrane with the x^1 axis [?], defined by

$$e_1^1 = \nu \cos\psi, \tag{102}$$
$$e_1^2 = -\nu \sin\psi. \tag{103}$$

Finally, the Helfrich membrane metric tensor is defined by [?]

$$g_{ij} \equiv \boldsymbol{e}_i \cdot \boldsymbol{e}_j. \tag{104}$$

In what follows, latin indices $i, j, k, \cdots$ will be used to denote vectors and tensors related to the Helfrich membrane manifold; their position as upper and lower indexes is thus important [?]. Because the differential manifold of Helfrich membrane is one dimensional, indexes $i, j, k, \cdots$ may take only one value, i.e., $i = j = k = \cdots = 1$. However, in formal relations such as ??, we will use the general notation $i, j, k, \cdots$ for the sake of generality, and for potential extensions of such relations to two-dimensional manifolds [?].

In order to work out the dynamical equations for Helfrich membrane, let us derive the force exerted by bulk fluid on Helfrich membrane. The force exerted on a line element $\mathrm{d}l_{\rm c}$ of Helfrich membrane is [?]

$$f_{{\rm F}\,\alpha}(\sigma_{\rm Fc}, \nu, \psi) = -\sigma^{\rm c}_{{\rm F}\,\alpha\beta}\hat{n}^{{\rm c}\,\gamma}(\nu, \psi), \tag{105}$$

where the bulk fluid stress tensor is evaluated at the current coordinate $\mathbf{x}_{\rm c}^{\rm M} \in \partial\Omega^{\rm c}_{\square}$ corresponding to x^1. The components of the force ?? tangential and normal to Helfrich membrane are [?]

$$f^i_{{\rm F}\,t}(\sigma_{\rm Fc}, \nu, \psi) = g^{ij}(\nu) e^{\alpha}_j(\nu, \psi) f_\alpha(\sigma_{\rm Fc}, \nu, \psi) \tag{106}$$

$$f_{{\rm F}\,n}(\sigma_{\rm Fc}, \nu, \psi) = \hat{n}^{{\rm c}\,\alpha}(\nu, \psi) f_\alpha(\sigma_{\rm Fc}, \nu, \psi), \tag{107}$$

where in ?? we used the feature that the metric tensor depends on ν only, see ????.

The equations of motion for Helfrich membrane are [? ? ?]

$$\nabla_i v^i_{\mathrm{M}} - 2H w_{\mathrm{M}} = 0, \tag{108}$$

$$\rho_{\mathrm{M}} \left(\frac{\partial v^i_{\mathrm{M}}}{\partial t} + v^j_{\mathrm{M}} \nabla_j v^i_{\mathrm{M}} - 2 v^j_{\mathrm{M}} w_{\mathrm{M}}\, b^i_j - w_{\mathrm{M}} \nabla^i w_{\mathrm{M}} \right) = \nabla^i \sigma_{\mathrm{M}} + f^i_{\eta_{\mathrm{M}}\, t} + f^i_{\mathrm{F}\, t}, \tag{109}$$

$$\rho_{\mathrm{M}} \left[\frac{\partial w_{\mathrm{M}}}{\partial t} + v^i_{\mathrm{M}} (v^j_{\mathrm{M}} b_{ji} + \nabla_i w_{\mathrm{M}}) \right] = 2\sigma_{\mathrm{M}} H + f_\kappa + f_{\eta_{\mathrm{M}}\, n} + f_{\mathrm{F}\, n}, \tag{110}$$

$$\frac{\partial \boldsymbol{u}_{\mathrm{M}}}{\partial t} = w_{\mathrm{M}} \hat{n}^{\mathrm{c}}, \tag{111}$$

$$\frac{\partial u^1_{\mathrm{M}}}{\partial x^1} = -1 + \nu \cos\psi, \tag{112}$$

$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = -\nu \sin\psi, \tag{113}$$

and they hold for $x^1 \in \partial\Omega^{\mathrm{r}}_{-}$. Here, v^i_{M} and w are the Helfrich membrane tangential and normal velocity, respectively, ∇ is the covariant derivative related to the Levi-Civita connection associated to g, ∇_{LB} the Laplace-Beltrami operator, b the second fundamental form, and H and K the mean and Gaussian curvature, respectively. We refer to [? ? ? ?] for details on the differential-geometry framework, and we will now discuss each line in ????????????. ?? enforces mass conservation [?]. ???? are the tangential and normal components of the Navier-Stokes equations, respectively. The first term in the right-hand side of ?? represnts the force on membrane elements induced by gradients of line tension, while the first term in the right-hand side of ?? constitutes Laplace pressure [?]. The other terms read

$$f_\kappa(\nu, \psi) \equiv -2\kappa[\nabla_i \nabla^i H + 2H(H^2 - K)], \tag{114}$$

$$f^i_{\eta_{\mathrm{M}}\, t}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv \eta_{\mathrm{M}}[-\nabla_{\mathrm{LB}} v^i_{\mathrm{M}} - 2\left(b^{ij} - 2\,H\,g^{ij}\right) \nabla_j w_{\mathrm{M}} + 2K v^i_{\mathrm{M}}], \tag{115}$$

$$f_{\eta_{\mathrm{M}}\, n}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv 2\eta_{\mathrm{M}}[(\nabla^i v^j_{\mathrm{M}}) b_{ij} - 2 w_{\mathrm{M}}(2H^2 - K)]. \tag{116}$$

In ??, f_κ is represents the resistance of Helfrich membrane to bending, and it is derived from Helfrich model [?], while $f_{\eta_{\mathrm{M}}\, t}$ and $f_{\eta_{\mathrm{M}}\, n}$ in ???? are, respectively, the tangential and normal component of the viscous force resulting from in-membrane flows [?]. ?? is the time-evolution equation for the Helfrich membrane shape, which evolves according to the normal $\hat{n}^{\mathrm{c}}$, and it has been obtained from ????. ???? re-express the definitions of ν and ψ, and they result from ??????????. Finally, ρ_{M}, σ_{M} and κ are, respectively, the density, line tension and bending rigidity of Helfrich membrane.

We choose the following boundary conditions for ????????????:

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{117}$$

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{118}$$

$$w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{119}$$

$$\nabla_i w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{120}$$

$$\sigma_{\mathrm{M}} = \sigma_{\mathrm{M}*} \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{121}$$

$$\boldsymbol{u}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{122}$$

$$u^1_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{123}$$

$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}. \tag{124}$$

?????? enforce the condition that the left extremity of Helfrich membrane, $\partial\Omega_{\bullet-}$, is clamped on the channel wall $\partial\Omega_{\square}$. ???????? enforce the mirror symmetry about $\partial\Omega_{\square}$. Namely, ?? ensures that the v_{M} vector field vanishes on $\partial\Omega_{-\bullet}$, because a nonzero value of this field on $\partial\Omega_{\square}$ would break the axial symmetry. The condition ?? follows the same lines. In addition, the symmetry requires that scalar fields and vertical components of vector fields in Ω have zero slope with respect to x^1 at $\partial\Omega_{-\bullet}$, see ????. Finally, ?? fixes the Helfrich membrane tension on $\partial\Omega_{\bullet-}$.

B. Bulk fluid domain motion

The equation of motion for bulk fluid domain are ????, with boundary conditions

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}, \tag{125}$$

$$u^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{126}$$

$$\frac{\partial u^2_{\mathrm{D}}}{\partial \mathrm{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{127}$$

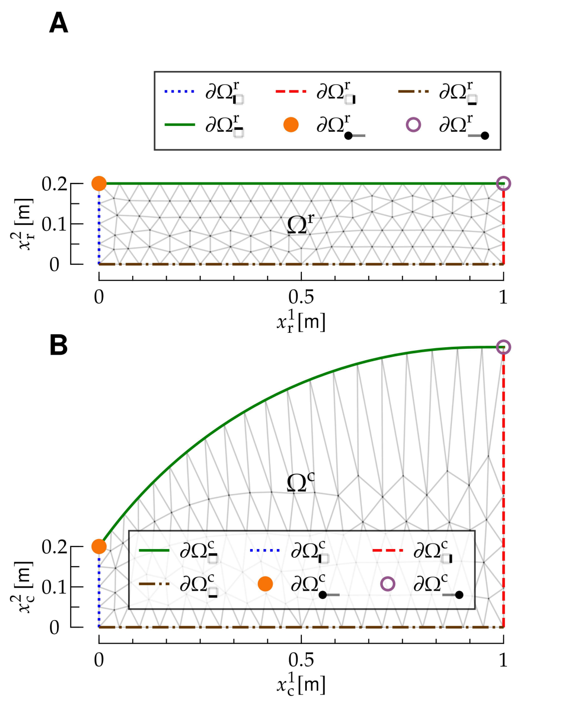

FIG. 6: Reference and current configuration for the interaction between a bulk fluid (F) and a Helfrich membrane (M). The figure follows the same notation as ??. The top edge of the bulk fluid mesh, $\partial\Omega^{\mathrm{r}}_{\square}$, coincides with the one-dimensional manifold $\Omega^{\mathrm{r}}_{\_}$ of Helfrich membrane.

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{128}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}, \tag{129}$$

$$\dot{u}^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{130}$$

$$\frac{\partial \dot{u}^2_{\mathrm{D}}}{\partial \mathrm{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{131}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \dot{\boldsymbol{u}}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{132}$$

?? ensures that bulk fluid domain is not deformed on $\partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}$, cf. **??**. **????** follow from axial symmetry, and **??** from the requirement that bulk fluid domain and Helfrich membrane deformations stay conforming [**?**]. Finally, the boundary conditions for $\dot{\boldsymbol{u}}_{\mathrm{M}}$, **????????**, are derived along the same lines as **????????**.

C. Bulk fluid motion

The bulk fluid equations of motion are **????**, with boundary conditions

$$\boldsymbol{v}_{\mathrm{F\,c}} = \boldsymbol{v}_{*} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{133}$$

$$\boldsymbol{v}_{\mathrm{F\,c}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{134}$$

$$v_{\mathrm{F}}{}^1_{\mathrm{c}} = 0 \;\text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{135}$$

$$\frac{\partial v_{\mathrm{F}}{}^2_{\mathrm{c}}}{\partial \mathrm{x}^2_{\mathrm{c}}} = 0 \;\text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{136}$$

$$\boldsymbol{v}_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{M}}(x^1)}{\mathrm{d}t}\right|_{x^1=[\boldsymbol{\psi}^t(\mathbf{x}_{\mathrm{c}})]^1} \quad \text{on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{137}$$

$$\sigma_{\mathrm{F\,c}} = \sigma_{\mathrm{F}\,*} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}. \tag{138}$$

According to **??**, bulk fluid is injected from the bottom edge of Ω^{c}, while **??** is a no-slip boundary condition on the left wall $\partial\Omega^{\mathrm{c}}_{\square}$, and **????** result from symmetry. **??** results from the conforming motion [**?**] of bulk fluid and Helfrich membrane. Finally, **??** stems from the presence of the reservoir, in which the bulk fluid tension is equal to $\sigma_{\mathrm{F}\,*}$: At the boundary $\partial\Omega^{\mathrm{c}}_{\square}$ between the channel and the reservoir, the bulk fluid tension matches the reservoir's.

The dynamics of the system is obtained as follows; the derivation is obtained along the lines of **????**, and details can be obtained by referring to those Sections.

Given the Helfrich membrane fields $v^{n-1,i}_{\mathrm{M}}$, $w^{n-1,i}_{\mathrm{M}}$, $\sigma_{\mathrm{M}}{}^{n-3/2}$, $\boldsymbol{u}^{n-3/2}_{\mathrm{M}}$, $\nu^{n-3/2}$, $\psi^{n-3/2}$, the bulk fluid domain fields $\boldsymbol{u}^{n}_{\mathrm{D}}$, $\boldsymbol{u}^{n-1}_{\mathrm{D}}$, and the bulk fluid fields $\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}$, $\sigma_{\mathrm{F\,r}}{}^{n-3/2}$, we determine the fields at the next time step as follows:

- **Update Helfrich membrane.** Solve the boundary-value problem

$$\rho_{\mathrm{M}}\Bigg[\frac{\overline{v}^i_{\mathrm{M}} - v^{n-1,\,i}_{\mathrm{M}}}{\Delta t} + \left(\frac{3}{2}v^{n-1,\,j}_{\mathrm{M}} - \frac{1}{2}v^{n-2,\,j}_{\mathrm{M}}\right)\nabla^{n-1/2}_j V^i_{\mathrm{M}} - 2V^j_{\mathrm{M}} W_{\mathrm{M}} b^{n-1/2,\,i}{}_j - W_{\mathrm{M}}\nabla^{n-1/2,\,i} W_{\mathrm{M}}\Bigg] = \nabla^{n-1/2,\,i}\sigma_{\mathrm{M}}{}^{*} + f^i_{\eta\mathrm{M}}\left(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2}\right) + \tag{139}$$

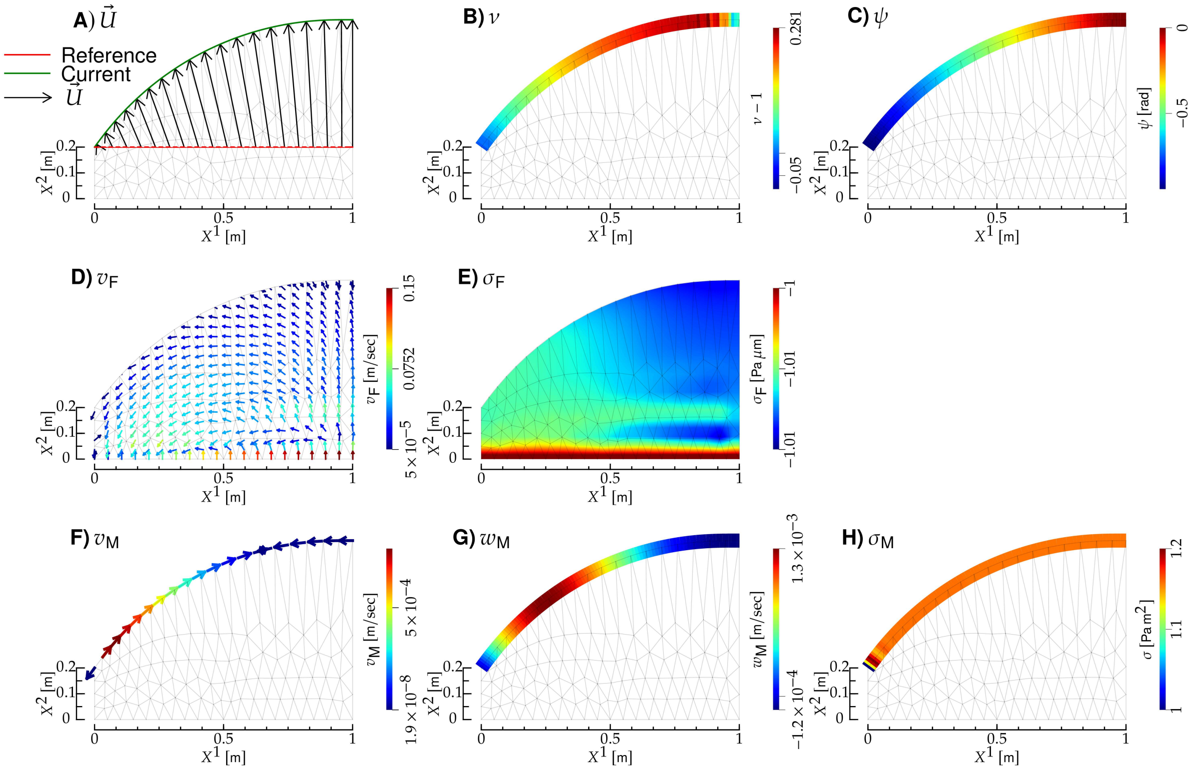

FIG. 7: Interaction between a bulk fluid and a Helfrich membrane. **A**) Membrane reference and current configuration (red and green curve, respectively), and displacement field (black arrows). **B**) Membrane stretch field. **C**) Membrane tangent angle. **D**) and **E**)) Bulk-fluid velocity and tension; the notation is the same as in **??**. **F**), **G**) and **H**): membrane tangential velocity, normal velocity and tension; the notation is the same as in ? B, C and D, respectively. All panels refer to the same instant of time, shown on top, and display also the deformed mesh (gray lines).

$$f^i_{\mathrm{F}\,t}(\boldsymbol{\sigma}_{\mathrm{F\,r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \sigma_{\mathrm{F\,r}}{}^{n-3/2}, \boldsymbol{u}_{\mathrm{D}}^{n-1}), \nu^{n-1/2}, \psi^{n-1/2}),$$

$$\rho_{\mathrm{M}}\left[\frac{\overline{w}_{\mathrm{M}} - w_{\mathrm{M}}^{n-1}}{\Delta t} + V^i_{\mathrm{M}} V^j_{\mathrm{M}} b^{n-1/2,\,j}{}_{i} + \left(\frac{3}{2} v_{\mathrm{M}}^{n-1,\,i} - \frac{1}{2} v_{\mathrm{M}}^{n-3/2,\,i}\right) \nabla_i^{n-1/2} W_{\mathrm{M}}\right] = \tag{140}$$

$$f_\kappa(\nu^{n-1/2}, \psi^{n-1/2}) + 2\sigma_{\mathrm{M}}{}^* H^{n-1/2} + f_{\eta_{\mathrm{M}}\,n}(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2}) +$$
$$f_{\mathrm{F}\,n}(\boldsymbol{\sigma}_{\mathrm{F\,r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \sigma_{\mathrm{F\,r}}{}^{n-3/2}, \boldsymbol{u}_{\mathrm{D}}^{n-1}), \nu^{n-1/2}, \psi^{n-1/2}), \tag{141}$$

$$\nabla_i^{n-1/2} \nabla^{n-1/2,\,i} \phi_{\mathrm{M}} = \frac{\rho_{\mathrm{M}}}{\Delta t}(\nabla_i^{n-1/2} \overline{v}^i_{\mathrm{M}} - 2H^{n-1/2}\overline{w}_{\mathrm{M}}), \tag{142}$$

$$\rho_{\mathrm{M}} \frac{\overline{v}^i_{\mathrm{M}} - v^{n,\,i}_{\mathrm{M}}}{\Delta t} = -\nabla^{n-1/2,\,i} \phi_{\mathrm{M}}, \tag{143}$$

$$w^n_{\mathrm{M}} = \overline{w}_{\mathrm{M}}, \tag{144}$$

$$\frac{u_{\mathrm{M}}^{n-1/2,\,\alpha} - u_{\mathrm{M}}^{n-3/2,\,\alpha}}{\Delta t} = w_{\mathrm{M}}^{n-1} \hat{n}^{\mathrm{c}\,n-1/2,\,\alpha}, \tag{145}$$

$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,1}}{\partial x^1} = -1 + \nu^{n-1/2} \cos \psi^{n-1/2}, \tag{146}$$

$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1} = -\nu^{n-1/2} \sin \psi^{n-1/2}, \tag{147}$$

with boundary conditions

$$\overline{v}^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{148}$$

$$\overline{v}^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{149}$$

$$\overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{150}$$

$$\nabla_i^{n-1/2}\overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{151}$$

$$\phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{152}$$

$$n_{\mathrm{M}}{}_i^{n-1/2}\nabla^{n-1/2,\,i}\phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-}, \tag{153}$$

$$\boldsymbol{u}_{\mathrm{M}}^{n-1/2} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{154}$$

$$u_{\mathrm{M}}^{n-1/2,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{155}$$

$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{156}$$

In **??**, n_{M} is the unit normal to $\partial\Omega_{-}$: It lies in the tangent bundle of Ω_{-}, is directed outside Ω_{-}, and is normalized according to

$$n_{\mathrm{M}}{}^i n_{\mathrm{M}i} = 1 \text{ [\textbf{? ? ?}]}.$$

- **Update bulk fluid domain.**
- **Update bulk fluid.**

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